ECM3412

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

COMPUTER SCIENCE

MAY 2013

Nature Inspired Computation

Module Leader: Dr Edward Keedwell

Duration: **2 HOURS**

Answer Question 1, and any two of the **THREE** other questions.
Question 1, is worth **40 marks**.
Other Questions are worth **30 marks** each.

Candidates are advised to spend **FORTY-FIVE** minutes on Question 1,
And **THIRTY-FIVE** minutes on other questions.

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examinations.

This is a CLOSED **BOOK** examination.

Compulsory question

1. (a) Write pseudocode for *an evolutionary algorithm* (EA). In your answer, explain how the main operations of the algorithm allow it to find good solutions within reasonable time.

(8 marks)

(b) Explain what is meant by a local maximum in a search landscape and why *hillclimbing algorithms* are unable to escape them.

(6 marks)

(c) Ant colony optimisation algorithms require a selection method to determine which path is selected at a particular junction in the construction graph.

(i) Briefly explain why this selection method should not always select the path with most pheromone on.
(ii) The path selection method usually contains two parameters α and β. Explain the purpose of these parameters and the effect they have on the behaviour of the search process.

(8 marks)

(d) Briefly describe three desirable features of the human immune system that would also be beneficial properties of an artificial immune system.

(6 marks)

(e) Explain the following terms as they relate to particle swarm optimisation.

i. Cognitive Component.
ii. Social Component.

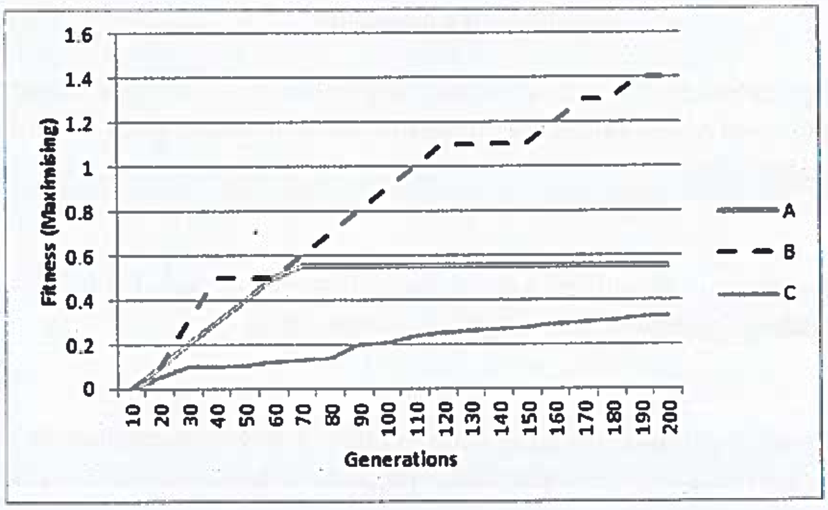
(4 marks)

(f) Explain the difference between *supervised* and *unsupervised* learning as they relate to the operation of neural networks. For each type of learning, illustrate your answer with an example of a neural network that uses this type of learning and the data that would be used to train it.

(8 marks)

(Total 40 marks)

Please Turn Over



2. (a) The figure above shows optimisation runs of three evolutionary algorithms, A,B and C plotting the fitness of the best solution through time. For each of the following, indicate which algorithm best fits the description and explain why you have made this decision.

i. An algorithm that has suffered from early convergence.

(5 marks)

ii. An algorithm that has low selection pressure.

(5 marks)

iii. An algorithm that has good selection pressure.

(5 marks)

(b) Crossover operations in evolutionary algorithms can sometimes 'break' a solution and require a 'fix' to enable the resulting children to solve the problem effectively, for example in the Travelling Salesman Problem (TSP). Answer the following, using diagrams where necessary.

i. Show how crossover can 'break' a solution to the TSP and explain why the solution is no longer valid for the problem.

(7 marks)

ii. Show how a solution can be 'fixed' by using the Partially Matched Crossover (PMX) operator.

(8 marks)

(Total 30 marks)

Please Turn Over

3. Ant colony optimisation is a successful algorithm that can be used for a large number of optimisation problems, including those that do not have an explicit topology. One such problem is that of water distribution optimisation where each pipe from the set of pipes p needs to be assigned a single diameter chosen from the finite set d. In this problem the network is simulated (using a hydraulic simulator) and a pressure value (in feet) is given for each node, this pressure must be equal to or greater than 15.0 for each node in the network. The cost of each pipe is calculated as *pipeDiameter*pipeLength*.

(a) Draw the *construction graph* needed to solve this problem using ant colony optimisation.

(8 marks)

(b) Explain how you would calculate the objective function to find a least cost network that meets or exceeds the pressure requirements.

(8 marks)

(c) Describe the parameter settings you would initially use to solve the problem using ant colony optimisation.

(8 marks)

(d) Describe the experiments you would undertake to determine whether your initial assumptions about the parameters are correct.

(6 marks)

(Total 30 marks)

4. Neural networks are nature-inspired algorithms that can be used for a variety of data processing and classification tasks.

   (a) Neural networks have a number of properties that make them similar to human brains. For each of the following, explain what is meant by each property in neural networks and its parallel in human brain physiology or behaviour.

      i. Connectionism.

      (4 marks)

      ii. Graceful degradation.

      (4 marks)

      iii. Generalisability.

      (4 marks)

   (b) Explain what is meant by an *activation function* in a neural network and give an example of a commonly used activation function.

   (3 marks)

   (c) Explain, with the aid of equations, how a multilayer perceptron

      i. calculates an output given a set of input data;

      (7 marks)

      ii. calculates the error on the output and updates the weights to generate a closer output in the next iteration.

      (8 marks)

(Total 30 marks)

End of Paper